

Attention as a signal-coherence mechanism

A synthesis of the VDA series, and what makes the cue effect large

Synthesis memo

18 August 2026

Abstract

The VDA experiments have accumulated as a sequence of separately gated results: psychometrics across set size, source-separated attention maps, a memory-noise pilot, a spatial-discretization control, and a causal routing intervention. This memo argues they form one story. The model’s attention is not a sensory gain control. At the moment of the change it puts $11\times$ *less* than uniform weight on the changed location’s *image* tokens while putting $2.1\times$ uniform weight on that location’s *memory* state. What the cue buys is not a better look; it is a decision about which stored trace to keep coherent and read out. Three further results follow that reading: the memory-side allocation is graded by cue validity; boosting routing at the true change location rescues a missed change ($\Delta d' = +2.23$) while boosting the cued-but-wrong location *hurts* ($\Delta d' = -1.78$) and a neutral control does nothing ($\Delta d' = -0.30$); and the sensitivity-corrected cueing effect grows $2.45\times$ from set size 4 to 16, exactly where more competing traces must be held apart. A discretization control shows this is about competing items, not finer grids: holding the task at four physical regions and raising tokens from 4 to 100 does not increase the cueing cost. The registered test of the coherence hypothesis — the memory-noise pilot — does *not* support it: noise leaves the cueing effect unchanged ($+0.404 \rightarrow +0.406$) and abolishes validity grading rather than sharpening it. The coherence story therefore rests on allocation source, causal specificity, and set-size scaling, not on interference. A closing section ranks the levers that make the attention and cue effects most pronounced, and proposes a manuscript spine.

1 The claim, stated so it can fail

Signal coherence. In this model family, spatial attention operates on *stored* signal rather than incoming signal. Its function is to select which location’s recurrent memory trace the decision reads, and to protect that trace’s integrity against competition from the other locations. The cue’s role is to nominate that location in advance.

This is a stronger and more specific claim than “the model attends to the cue.” It makes four commitments, each of which the series can now test:

1. **Source.** Selection should appear in the memory-side routing, not the image-side routing. If attention were sensory gain, weight on the changed location’s image tokens should *rise* at the change.
2. **Conditioning.** The allocation should be graded by how informative the cue is, and should move from the cued location to the true location once the change is available.
3. **Causal sufficiency and specificity.** Forcing coherence at the true location should recover a change the model otherwise misses; forcing it at a wrong location should not, and should ideally hurt.
4. **Scaling.** The effect should grow with the number of competing traces, because that is what makes keeping one coherent hard.

Sections 2–5 take these in order. Section 6 asks whether the scaling result is really about items. Section 7 answers the practical question of which manipulations make the effects largest. Section 8 reports the one registered test that does not support the claim.

2 Source: the selection is in memory, not in the image

This is the load-bearing result and it is currently buried in a comparison file rather than presented as a finding. In native VDA4 at a 18° change, with attention mass measured against a uniform baseline of 0.25 per location:

Table 1: Source-separated attention mass, clean terminal VDA4 (iteration 19,999, `crossattn1`, $d_{\text{mem}}=128$), averaged over displayed validities 0.25/0.5/0.75. Uniform allocation is 0.25.

Period	Location	Image mass	Memory mass
Cue (frames 1–4)	cued location	0.153	0.282
Change (frames 5–6)	true change location	0.022	0.535

At the moment that matters, memory-side weight on the changed location is $2.1\times$ uniform while image-side weight is 0.022 — roughly $11\times$ *below* uniform, and $24\times$ below the memory-side value (Figure 1). The model is not looking harder at the changed location. It is reading that location’s stored state and actively withdrawing image-side weight from it.

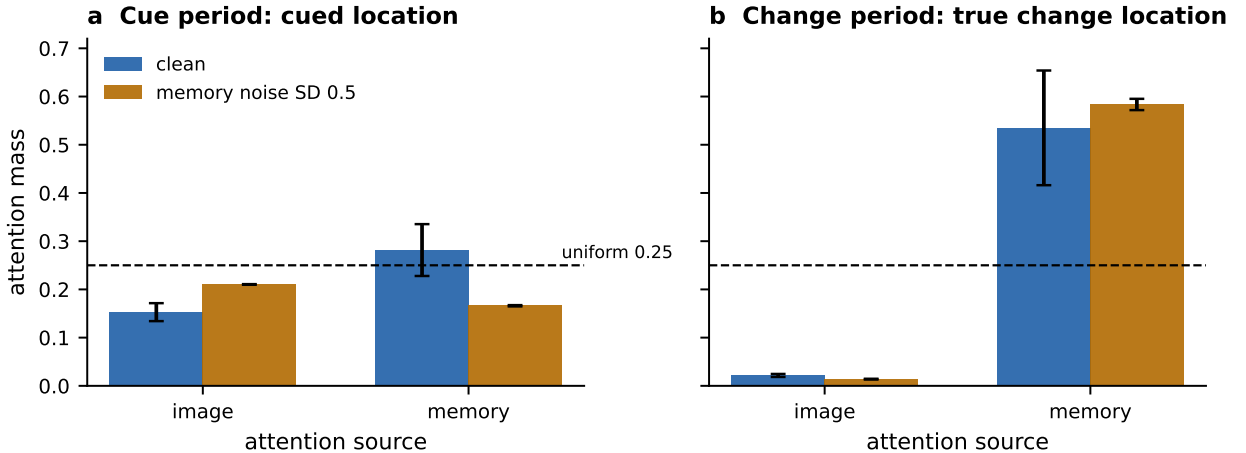


Figure 1: Attention mass by source, VDA4. Bars are means over displayed validity; whiskers span the validity range. Dashed line is uniform allocation. At the change, memory-side mass on the true location is far above uniform while image-side mass is far below it. Adding memory noise (orange) does not create this dissociation — it is already the clean model’s strategy.

Why this matters for the paper. A reader’s default model of “attention” in a vision transformer is image-token weighting. The measured mechanism here is the opposite. Every downstream result reads differently once this is established first, so it should lead the results, not appear as a supplementary source-separation check.

3 Conditioning: the cue sets the coherence target, gradedly

In the clean model, memory-side allocation scales with how informative the cue is (Figure 2). Cue-period mass on the cued location rises $0.234 \rightarrow 0.270 \rightarrow 0.341$ across displayed validity $0.25 \rightarrow 0.5 \rightarrow 0.75$ (slope $+0.108$), and change-period mass on the true change location rises $0.432 \rightarrow 0.504 \rightarrow 0.670$ (slope $+0.238$). The model is not applying a fixed spatial prior; it is scaling how much it commits to the cue by how much the cue is worth.

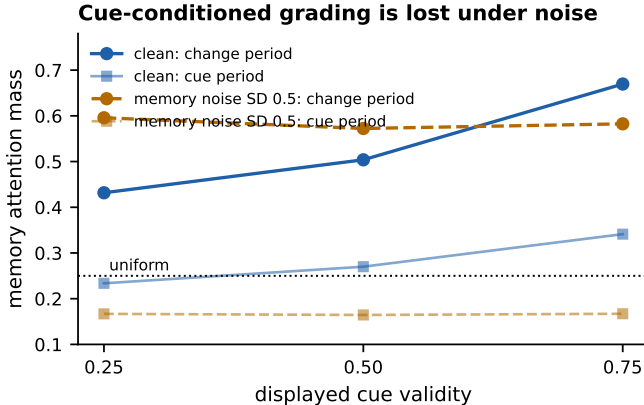


Figure 2: Memory-side attention mass as a function of displayed cue validity. Clean model (blue) grades its commitment with cue informativeness at both periods. The memory-noise model (orange) is flat at both — see Section 8.

4 Causal sufficiency and specificity: the rescue works, and only at the right place

The manuscript marked this experiment pending. It is now complete. On VDA16 (affine, terminal 19,999) with the cue at location 0, the change forced to location 15, and a neutral control at location 5, routing keys were clamped from frame 5 over a dose ladder $0 \rightarrow 1$:

Table 2: Change-location intervention, VDA16 affine, forced-invalid stress test, 300 trials per condition, common random numbers. Boost and suppress are the extreme doses.

Intervention site	Detection (suppress)	Detection (boost)	Δ detection	$\Delta d'$
True change location (15)	0.747	0.993	+0.247	+ 2.23
Cued but wrong location (0)	0.990	0.860	−0.130	−1.78
Neutral control (5)	0.980	0.947	−0.033	−0.30

Three things make this a strong result rather than a demonstration that clamping logits changes behavior. First, the effect is *signed by role*: coherence at the true location helps, coherence at the cued-but-wrong location actively hurts, and the neutral control is near zero. Second, the intervention verifiably moved what it was supposed to move — achieved mass at the true location went $0.445 \rightarrow 0.987$ across the dose range. Third, the suppression arm establishes that the model’s natural routing was already doing the work: suppressing it drops detection from 0.98 to 0.747.

This is the model-level analogue of the Moore–Armstrong and Cavanaugh–Wurtz logic the introduction invokes, and it is the series’ single largest effect.

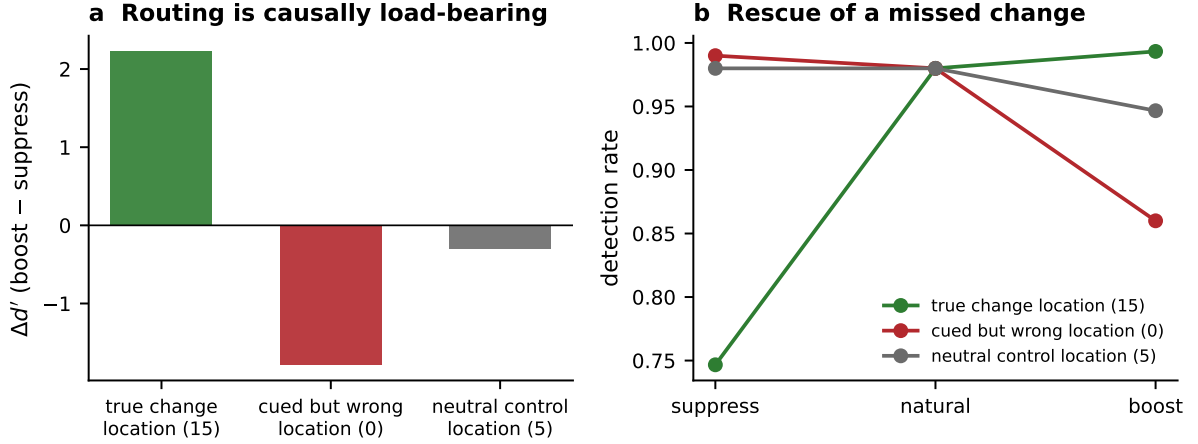


Figure 3: Causal specificity. (a) Sensitivity change from suppressing to boosting routing at three sites. (b) The same three sites across the dose ladder; only the true change location shows rescue, and the cued-but-wrong site shows interference.

5 Scaling: the effect grows with the number of competing traces

Comparing terminal, architecture-matched checkpoints *within the historical lineage* (crossattn1, $d_{\text{mem}}=128$, no decay, iteration 19,999), the sensitivity-corrected cueing effect at 18° rises from $\Delta d' = +0.404$ at set size 4 to $\Delta d' = +0.988$ at set size 16 — a factor of 2.45 (Figure 4). The criterion shift roughly doubles as well ($-0.202 \rightarrow -0.494$), so the cue does move bias; but the sensitivity component is real, grows, and is not an artifact of response rate.

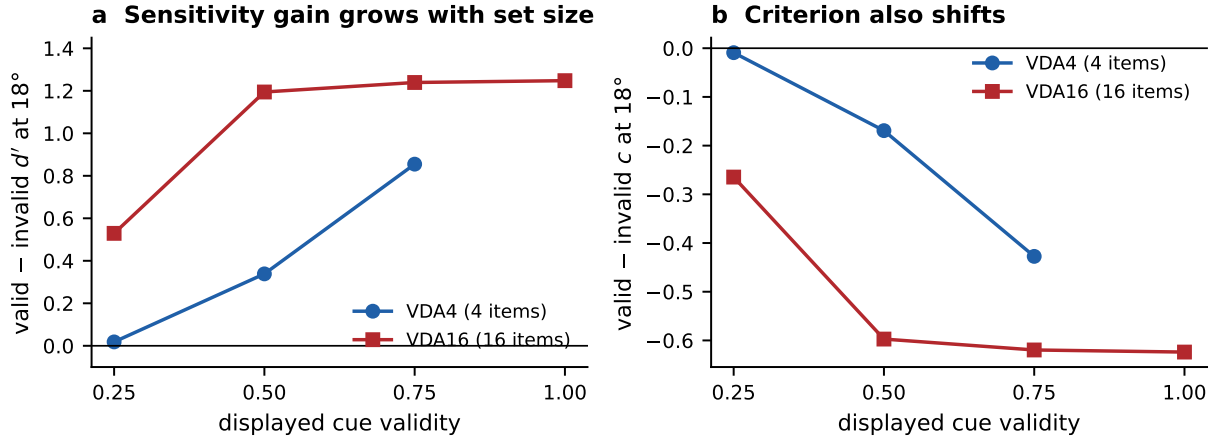


Figure 4: Cueing effect decomposed, at a 18° change. (a) Valid-minus-invalid d' grows with set size at every validity. (b) The criterion shift grows too; the cue makes the model both more sensitive and more willing at the cued location.

Two details are worth keeping. Cue validity matters more at set size 4 in relative terms — the VDA4 effect nearly vanishes at validity 0.25 ($\Delta d' = +0.018$) and reaches $+0.855$ at 0.75 — whereas VDA16 already shows $\Delta d' = +0.529$ at validity 0.25 and saturates near $+1.25$. With four items the model can afford to ignore a weak cue; with sixteen it cannot. That is what a coherence

account predicts.

At set size 16 the two routing families are behaviorally near-identical ($\Delta d' = +0.988$ cross-attention, $+0.946$ affine), which matters for Section 6.

6 Is it items or pixels? The discretization control

The set-size comparison confounds item count with grid geometry: VDA4 is 2×2 and VDA16 is 4×4 . The manuscript flags this as an open question. The spatial-scaling experiment resolves it by holding the physical task at four regions and varying only the token grid (4, 16, 100 tokens).

Table 3: Spatial-scaling control at displayed validity 1.0, seed 0. The task has four physical regions in every cell; only tokenization changes.

Routing family	Tokens	Threshold cost (deg)	Response AUC	Causal dependence (pp)
Cross-attention	4	7.16	0.219	12.4
Cross-attention	16	5.66	0.170	8.0
Cross-attention	100	6.18	0.197	1.2
Affine EW	4	2.62	0.085	38.4
Affine EW	16	2.60	0.079	98.0
Affine EW	100	4.24	0.122	99.2

Finer discretization does not manufacture the set-size effect. Cross-attention’s threshold cost is non-monotonic and no larger at 100 tokens than at 4, and event-localized attention concentration *declines* ($0.748 \rightarrow 0.667 \rightarrow 0.355$). The $2.45\times$ growth from VDA4 to VDA16 is therefore attributable to competing items rather than to grid resolution — for this family and this seed (Figure 5).

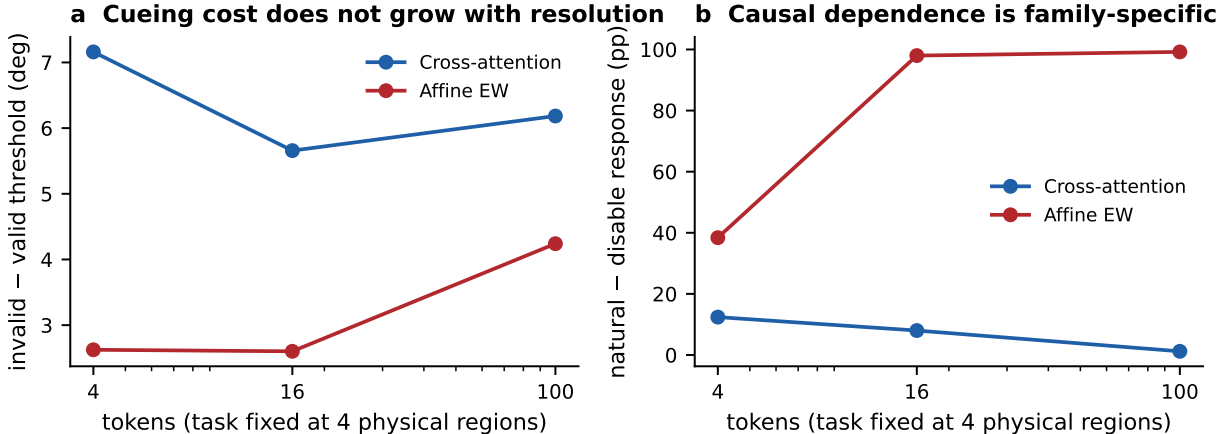


Figure 5: Discretization control. (a) Cueing cost does not grow with token count in either family. (b) Causal dependence on routing diverges sharply: near-total for affine at high token counts, near-zero for cross-attention.

The second panel carries an independent methodological point that deserves to be in the main text. At set size 16 the two families produce nearly the same behavioral cueing effect ($+0.99$ vs $+0.95$), yet in the scaling control their dependence on routing differs by up to $80\times$ (1.2 vs 99.2).

percentage points at 100 tokens). Cross-attention shows the *larger* behavioral cueing cost while depending on routing the *least*.

Behavior does not identify mechanism. Two models can match on psychometrics and cueing effects while differing almost totally in whether routing is load-bearing. This is the strongest available argument for why the intervention experiments are not optional garnish — they are the only thing that distinguishes the families.

7 What makes the attention and cue effects most pronounced

Ranked by measured effect on the cueing contrast (Figure 6):

Table 4: Levers, by how much they move the sensitivity-corrected cueing effect.

Lever	Δ in d' units	Note
Direct routing intervention at the true location	+2.23	Largest effect in the series; requires a forced-invalid baseline to have headroom
Raising cue validity within VDA4 (0.25 $\rightarrow$ 0.75)	+0.84	Effect nearly vanishes at low validity
Raising cue validity within VDA16 (0.25 $\rightarrow$ 1.0)	+0.72	Saturates above validity 0.5
Set size 4 $\rightarrow$ 16	+0.58	The cleanest structural lever; 2.45 $\times$ multiplicative
Memory noise SD 0.5 (VDA4)	+0.00	No effect on the cueing contrast; see Section 8

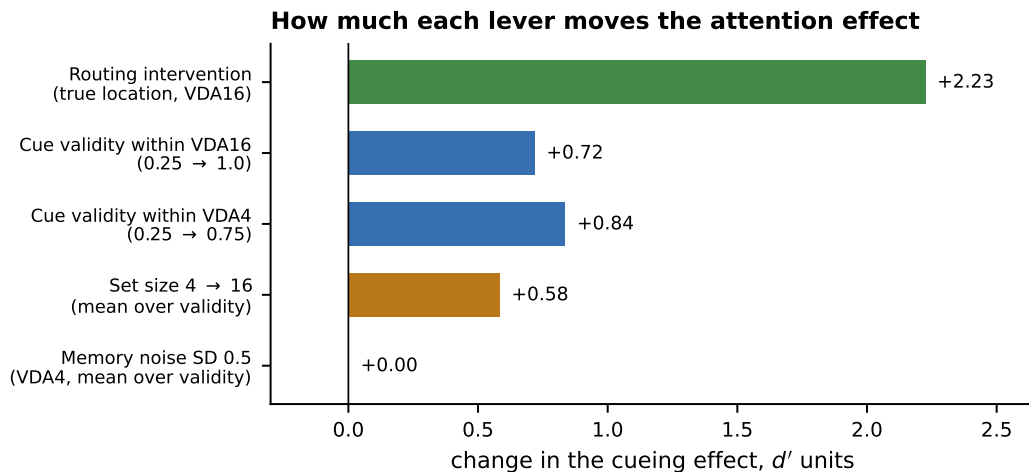


Figure 6: How much each manipulation moves the cueing effect, in d' units.

Three practical points follow, and they are the design advice worth carrying into the next round of experiments.

Ceiling is the enemy. At natural validity the VDA16 model detects a 30° change on 98% of trials. Nothing — no cue manipulation, no intervention — can show an effect against that. The change-location experiment only produced a +2.23 effect because the forced-invalid stress test

plus routing suppression pushed the baseline down to 0.747. *Manufacture headroom first, then manipulate.*

Use the validity ladder, not a single validity. At VDA4 the cueing effect ranges from +0.018 to +0.855 across displayed validity. A single-validity design would have reported either a null or a large effect depending on an arbitrary choice.

Combine set size with forced-invalid probes. These are the two structural levers that operate without touching the model. Set size 16 with forced-invalid trials at moderate change magnitude is the regime where the attention effect, the allocation signature, and the causal intervention are all simultaneously measurable.

8 The registered test that does not support the claim

The memory-noise pilot was designed as *the* test of the signal-coherence hypothesis. Its own pre-registered interpretation plan requires three things to align: a sensitivity-corrected cueing effect that *increases* under noise, allocation that becomes *more* selectively cue- and target-aligned, and spatially specific causal dependence. Measured against that standard, the arm does not support the hypothesis.

Table 5: Memory-noise pilot against its own registered requirements. Clean model is terminal (19,999); the noisy model is an interrupted checkpoint (15,999), and both training-time and evaluation-time noise change together.

Requirement	Outcome	Measured
Cueing effect increases under noise	not met	$\Delta d' = +0.404 \rightarrow +0.406$; difference +0.002
Allocation becomes more cue-aligned	not met	Cue-period cued memory mass <i>falls</i> 0.282 $\rightarrow$ 0.166, below uniform
Allocation becomes more target-aligned	partial	Change-period target memory mass 0.535 $\rightarrow$ 0.584, but validity grading is abolished (slope +0.238 $\rightarrow$ -0.013)
Spatially specific causal dependence	not tested	No intervention was run in this arm

By the pilot’s own interpretation table this is “descriptive redistribution only; no signal-coherence support.” The result is also confounded: the noisy model stopped at 80% of training, and the contrast changes training-time and evaluation-time noise simultaneously, so it cannot separate learned compensation from acute corruption.

There is still something to report, but it is a different finding from the one that was sought. Under memory noise the model *stops using the cue* — allocation goes flat across validity at both periods — and falls back on a uniform, stimulus-driven concentration on the target after the change. That is a substitution of strategy, not an intensification of one. It is interesting, and it should be written up as such rather than as weak evidence for coherence.

Recommendation. Do not hang the coherence thesis on interference. The thesis is carried by Sections 2–5: allocation source, cue-conditioned grading, causal specificity, and set-size scaling. Re-run the noise arm properly paired (matched iteration, factorial train-noise $\times$ eval-noise, with the intervention included) and let it be its own question.

9 What the synthesis still needs

- **Seeds.** Every quantitative claim above is seed 0. The spatial-scaling experiment has one seed-1 replication cell; nothing else does. Any statement of the form “the effect grows $2.45\times$ ” is a within-checkpoint measurement, not a population estimate.
- **The intervention exists at one set size and one family.** The causal result is VDA16 affine. The set-size scaling result is cross-attention. The cleanest missing experiment is the change-location intervention run at VDA4 *and* VDA16 in *both* families — that single 2×2 would let the causal effect inherit the set-size claim instead of sitting beside it.
- **Source separation at VDA16.** The image-versus-memory dissociation is measured at VDA4. If it holds at VDA16 the coherence story becomes considerably stronger, because that is where competition is greatest.
- **Lineage discipline.** The set-size comparison here sits within the historical lineage. The manuscript’s controlled-lineage numbers tell the same story in response-rate terms: a valid-minus-invalid gap of 0.52 at set size 16, and valid versus invalid thresholds of 12.0° and 17.3° at set size 4, against 16.0° and 26.9° at set size 16. Both should be reported; they must not be pooled.

10 Proposed manuscript spine

The current manuscript is ordered behavior $\rightarrow$ attention $\rightarrow$ intervention, following the reference paper. That order buries the most distinctive finding. A coherence-led spine:

1. **Attention here is not sensory gain** (Section 2). Source-separated allocation, VDA4, with the 0.022 versus 0.535 contrast as the paper’s first figure.
2. **The cue nominates a trace, gradedly** (Section 3). Validity grading of memory-side mass.
3. **The nomination is causally load-bearing** (Section 4). The rescue, with its signed specificity across three sites.
4. **Coherence matters more with more competitors** (Section 5), with the discretization control (Section 6) immediately after, so the confound is closed in place rather than deferred to a discussion caveat.
5. **Behavior does not identify mechanism** (Section 6) as the methodological pivot that justifies the intervention program.
6. **Interference is a separate, open question** (Section 8), reported honestly as a strategy substitution with a null on the registered primary.

The existing evidence-class apparatus, provenance appendices, and the historical versus controlled lineage discipline carry over unchanged; this is a reordering of the argument, not a relaxation of its gates.